

Apicomplexa

“**Apis**” are a lineage of **parasites** and **symbiotonts** to many animals. Though they need a host, they share ancestry with free-living photosynthetic algae.

Plasmodium falciparum (which causes malaria) and *Toxoplasma gondii* (which causes Toxoplasmosis) are both Apicomplexa that can infect people.

Toxoplasma can cause changes in the behavior of the mice they infect, making them less fearful of cats.

apical complex

This structure allows these parasites to enter their hosts

Dactylosoma kermi, an apicomplexan that infects frogs was named after Kermit the Frog!

mitochondrion

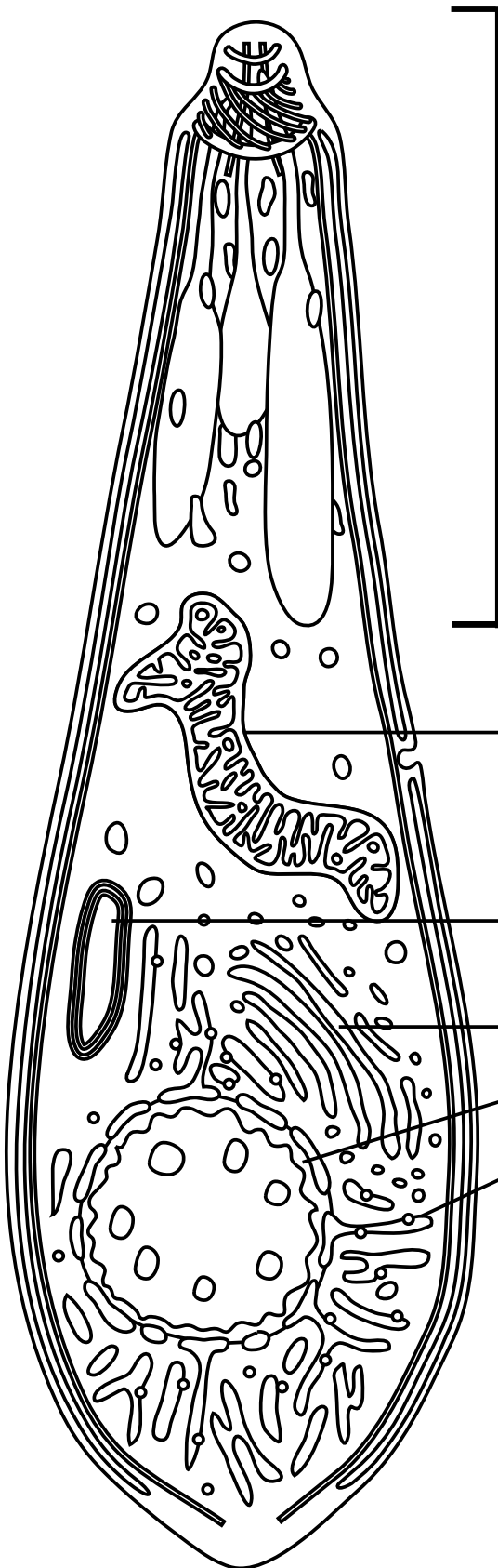
apicoplast

This structure is a chloroplast which has lost its function of photosynthesis. The ancestor of all Apicomplexa gained its **apicoplast** from a **Red Algae**

Golgi apparatus

nucleus

endoplasmic reticulum



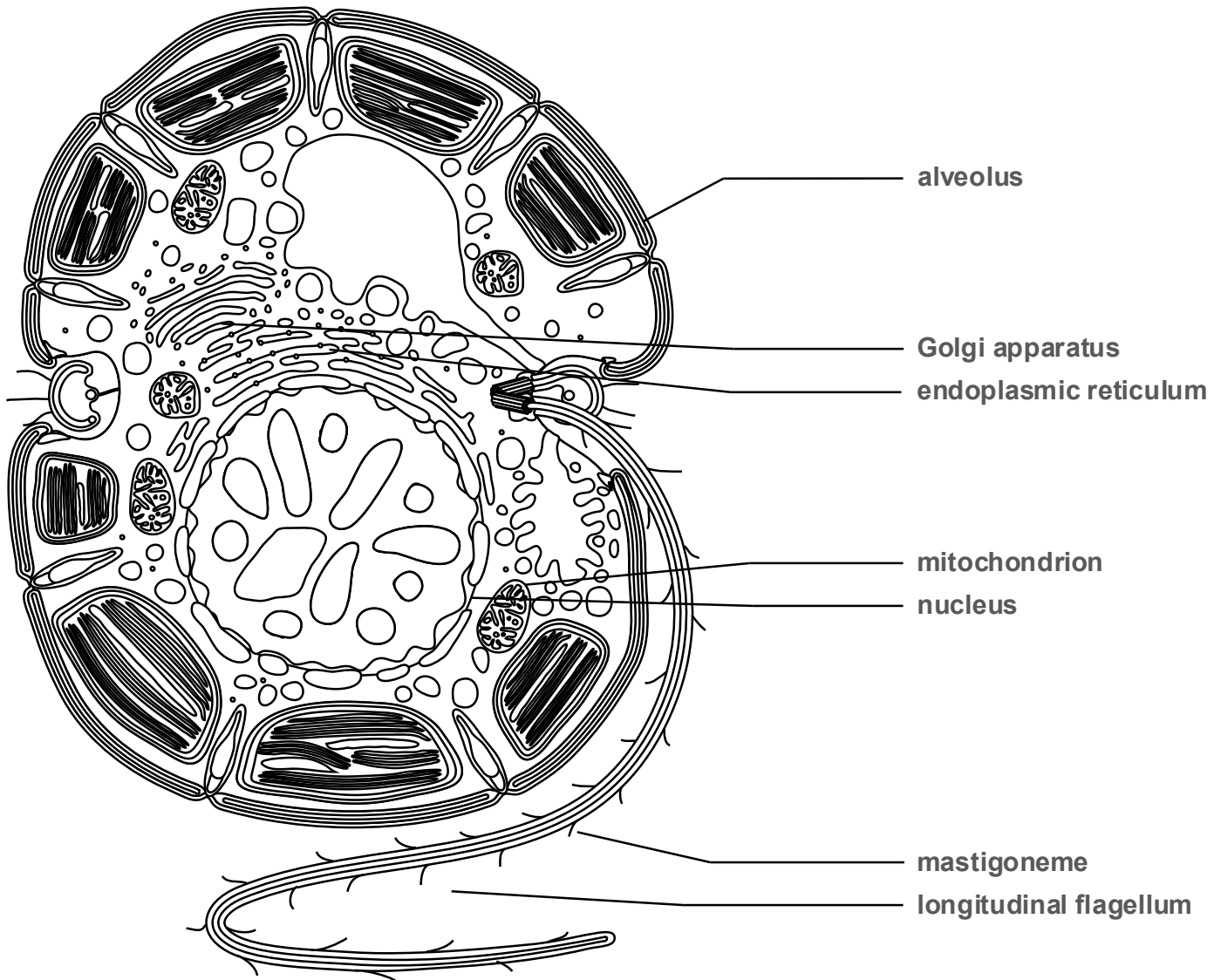
Dinoflagellate

“Dinos” live in marine and freshwater environments. Some are **mixotrophic** (photosynthetic AND capture prey), and some are completely non-photosynthetic. The photosynthetic taxa are important **phytoplankton** which produce oxygen.

Some species are **bioluminescent** and can be seen at night glowing in water, even here in Vancouver!

Some other species are toxic and can cause **red tide** when there is an algal bloom. And some dinoflagellates are parasitic to other organisms.

Zooxanthellae are dinoflagellate photosynthetic symbionts found in corals.



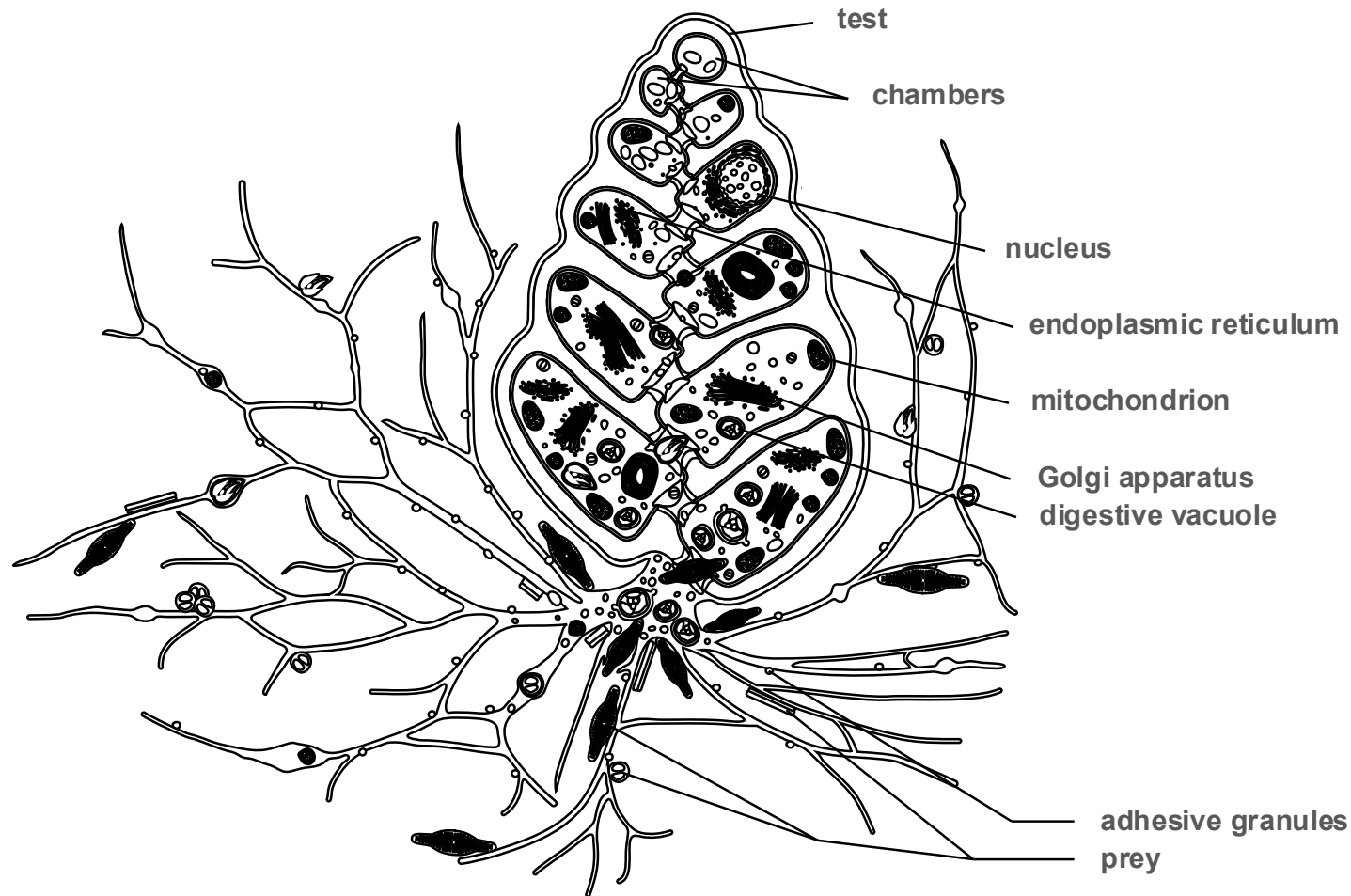
Foraminiferan

"Forams" are amoeboid predators of the ocean!

Baclogypsina sphaerulata is the species that make star sand (star-shaped **exoskeletons** which can be seen with the naked eye!)

Xenophyophorea is a **clade** of large, **multinucleated** foraminifera that look like sea sponges, but are actually one massive cell!

Tests are microscopic shells that can be made of a variety of materials based on the species. Some are secreted calcium carbonate, organic material, or can be **agglutinated** sand.

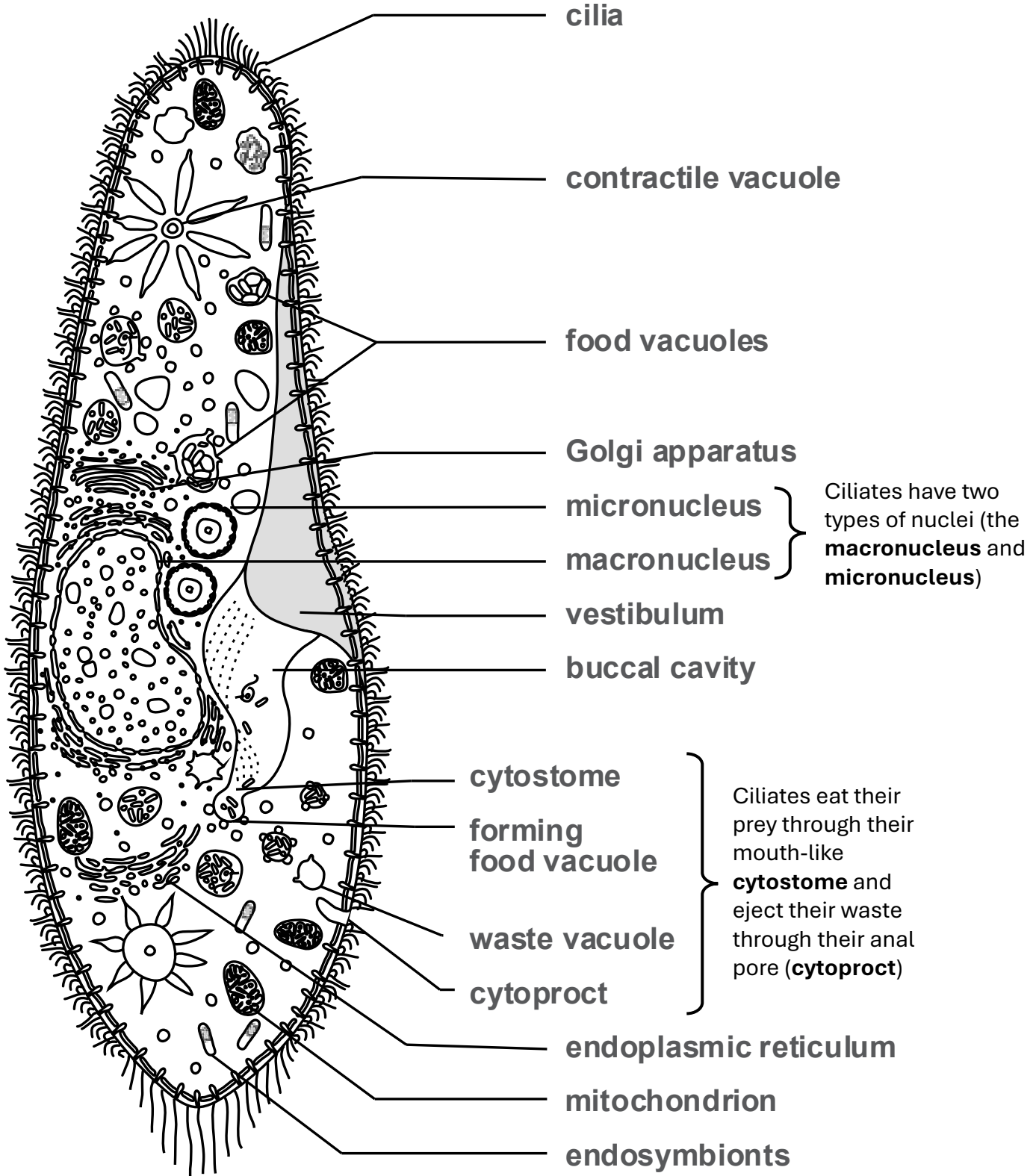


Forams use their long, sticky **pseudopodia** to capture prey.

Ciliate

Ciliates are predators of the microbial world and exhibit feeding behavior

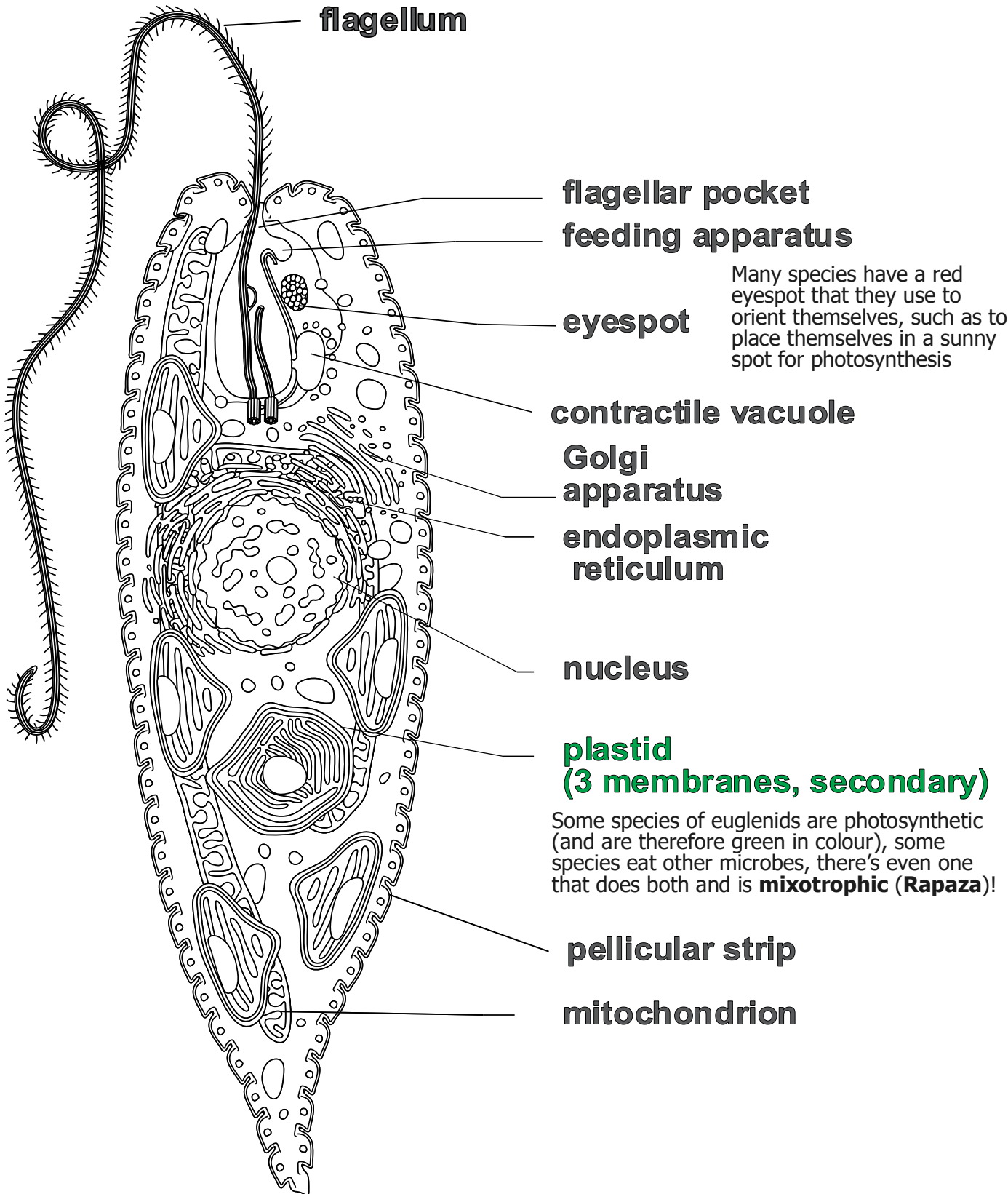
Some cells are large enough to see with the naked eye!



Euglenid

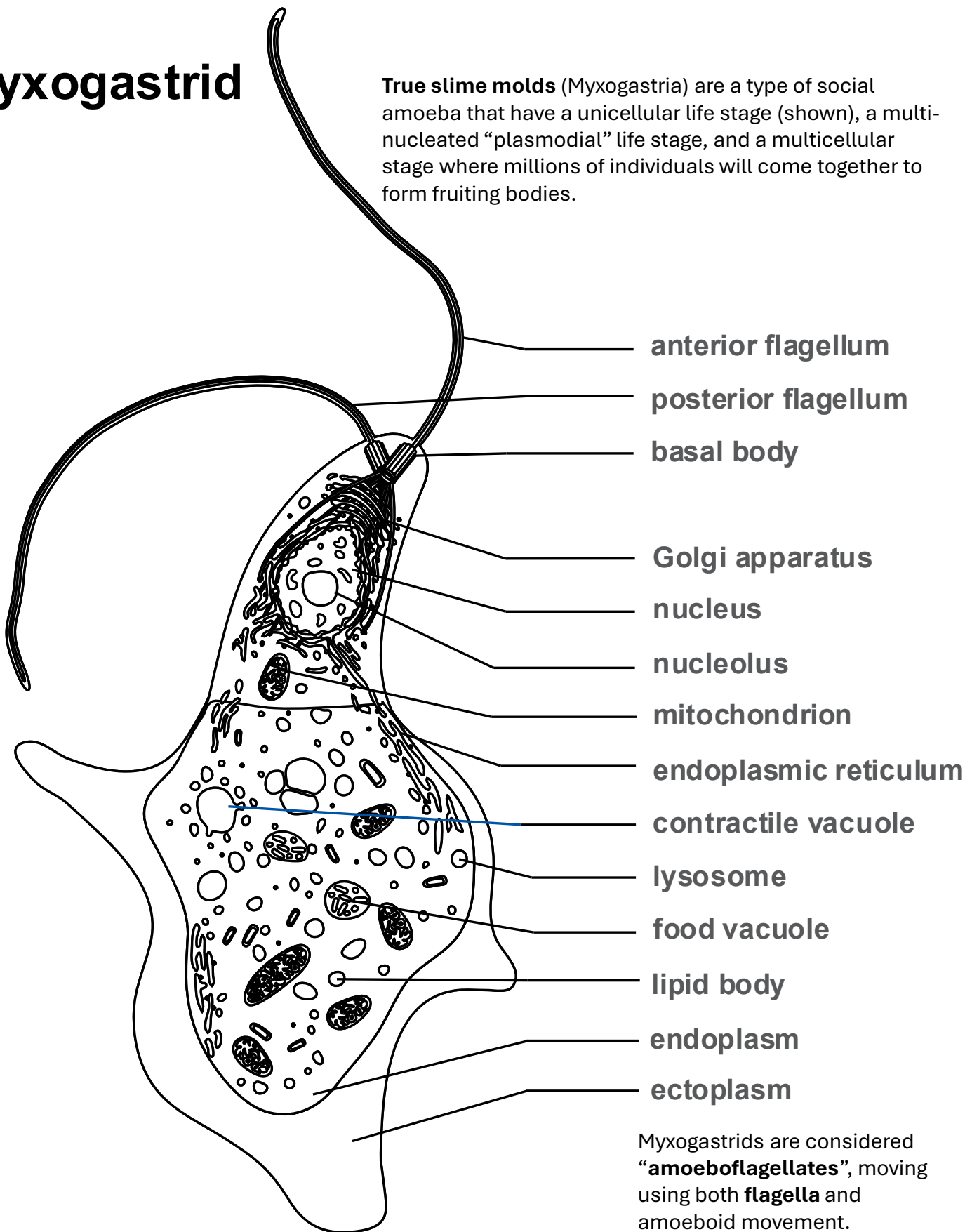
Euglenids can be found in freshwater, ocean, and the soil.

Many euglenids can change their shape quickly through 'euglenoid **metaboly**'. That means they can eat large food, like long diatoms.



Myxogastroid

True slime molds (Myxogastria) are a type of social amoeba that have a unicellular life stage (shown), a multi-nucleated “plasmodial” life stage, and a multicellular stage where millions of individuals will come together to form fruiting bodies.



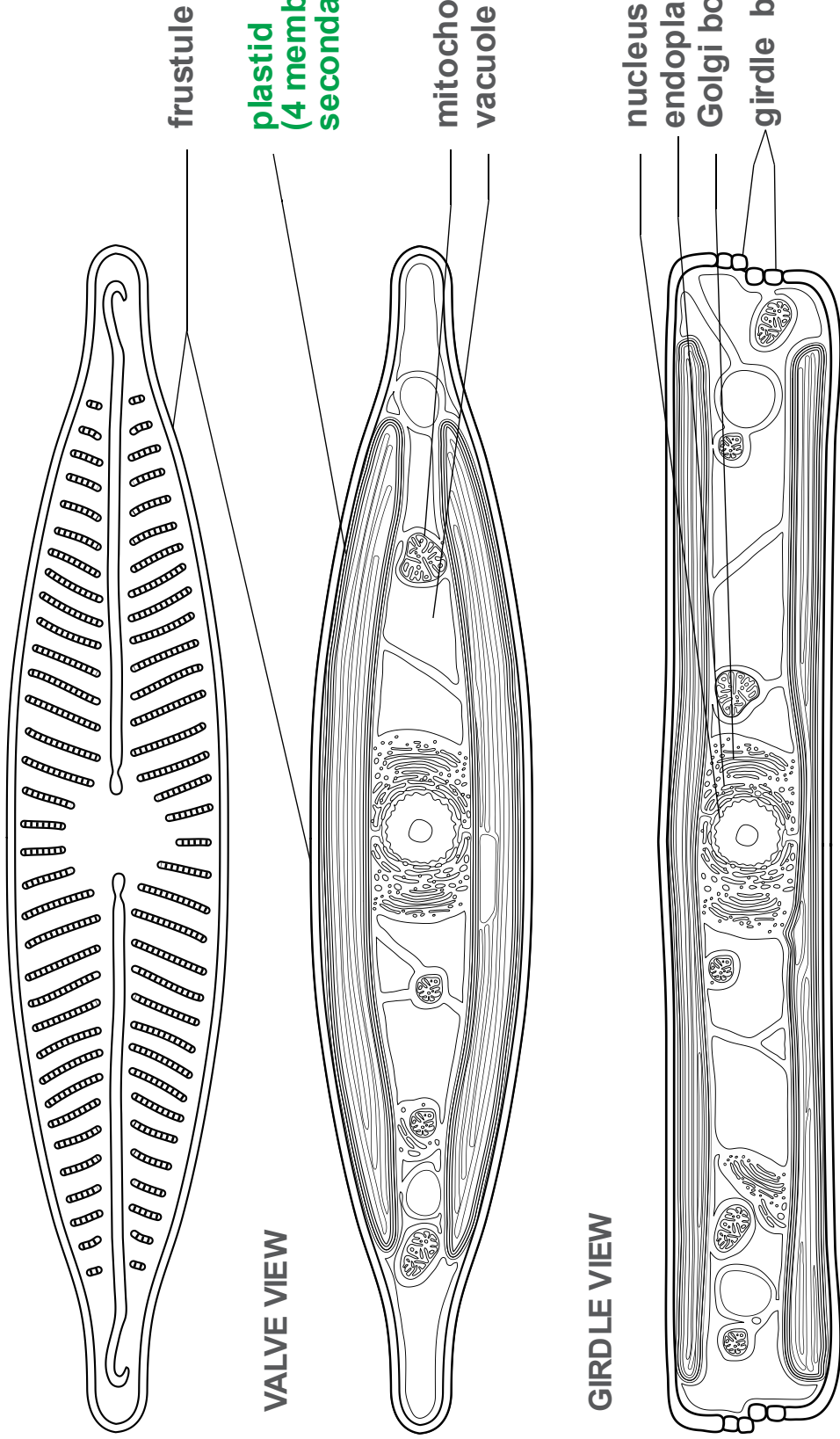
Myxogastroids are considered “**amoeboflagellates**”, moving using both **flagella** and amoeboid movement.

Diatoms are photosynthetic algae that have **silica** (glass) shells called **frustules**. These organisms are very common in marine and freshwater environments.

Diatoms come in a variety of shapes, sizes, and colors. Some people make art with Diatom frustules making complex mosaics.

Diatom

Diatomaceous earth is made from diatom frustules and used for pest control without chemicals.



Parabasalian

Parabasalialia live in anerobic environments. Their mitochondria have evolved into **hydrogenosomes** which generate ATP anaerobically (without oxygen)

Termites have parabasalian symbionts in their digestive tract to help with the digestion of wood.

